

GC-12: Spectral Engineering Predictions as Falsifiable GC Instances

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Abstract

BST's Spectral Engineering (SE) program produces laboratory-testable predictions. This note re-frames three SE predictions as formal instances of the Geometric Constraint (GC) method (GC-5: constraint/certificate/boundary), making their falsifiability structure explicit. Each prediction has a specific physical setup, a BST-derived outcome, a computational certificate, a laboratory verification protocol, and a stated boundary. Each can kill BST if the prediction fails. That is the point.

GC-11 cataloged eight engineering fields where geometric constraint is already the operative mechanism. This note narrows the focus: not “where does GC appear in engineering generally?” but “where does BST make a specific, falsifiable engineering prediction that follows the GC pattern?” The answer is three experiments, costing between \$10K and \$100K, executable with 2026 technology, each targeting a different BST integer.

1. Why Falsifiability Through GC

The GC method has three moves (GC-5 Section 1):

1. Constraint – Find independent bounds that pin the structure.
2. Certificate – Verify computationally that the constraint is tight.
3. Boundary – State what you did NOT prove.

For pure mathematics, the certificate is a proof. For engineering, the certificate must ultimately be a measurement. A prediction that cannot fail is not a prediction. BST's SE experiments are designed so that failure at the certificate step – a measurement that contradicts the BST-predicted outcome – falsifies a specific claim of the theory. No rescue, no re-interpretation, no “well, there might be corrections.” The prediction is sharp, the measurement is standard, and the outcome is binary.

This is what separates BST from theories that only post-dict known quantities. Post-diction is necessary (BST has 600+ confirmed post-dictions) but insufficient. Pre-diction that can fail is the gold standard.

2. Prediction 1: BaTiO₃ 137-Plane Piezoelectric Anomaly

BST's killer test. The one we want skeptics to run first.

2.1 Physical Setup

Fabricate a series of BaTiO₃ (barium titanate) thin films by pulsed laser deposition (PLD) on SrTiO₃ substrates. PLD achieves single-unit-cell thickness control, which is routine in oxide thin film laboratories worldwide. Prepare films at thicknesses from 125 to 150 unit cells (approximately 50-60 nm),

in increments of 1 unit cell. Measure the piezoelectric coefficient d_{33} of each film using a standard piezoresponse force microscope (PFM) or a Berlincourt meter.

The lattice constant of tetragonal BaTiO₃ is $a = 4.01$ Angstrom (c-axis). At 137 unit cells, the film thickness is 137×4.01 Angstrom = 549.4 Angstrom = 54.9 nm.

2.2 GC Constraint Specification

Two independent constraints pin the prediction:

Constraint 1 (Spectral cutoff). The D_{IV}^5 eigenvalue ladder has a spectral cap at $N_{\max} = 137$. This is not a free parameter – it is derived: $N_{\max} = N_c^3 * n_c + \text{rank} = 27 * 5 + 2 = 137$. The cap determines the maximum number of independent spectral modes that contribute coherently to any lattice-periodic physical process. At exactly $N_{\max}$ lattice planes, all 137 modes contribute constructively. Below $N_{\max}$, the sum is incomplete. Above $N_{\max}$, additional planes do not add new independent modes – they repeat.

Constraint 2 (Piezoelectric coupling). BaTiO₃ is a perovskite ferroelectric whose piezoelectric response couples to the lattice polarization. The polarization is a sum over phonon modes weighted by their contribution to the spontaneous polarization. The phonon spectrum is periodic in reciprocal space with period $2\pi/a$. The number of independent phonon branches contributing to d_{33} in the thin-film geometry is bounded by the film thickness in unit cells.

Independent bounds meet. Constraint 1 (from BST spectral geometry) and Constraint 2 (from condensed matter phonon physics) independently predict that the piezoelectric response as a function of film thickness should show an anomaly – a peak, kink, or saturation change – at exactly $N = 137$ unit cells. Constraint 1 says 137 is special because it is the spectral cap. Constraint 2 says the piezoelectric sum saturates when all independent phonon modes are included. BST says these are the same statement: the phonon mode count IS the spectral eigenvalue count.

2.3 BST-Predicted Outcome

The piezoelectric coefficient $d_{33}(N)$ as a function of film thickness N (in unit cells) shows an anomalous feature at $N = 137$. Specifically:

- For $N < 137$: d_{33} increases with N as more spectral modes contribute.
- At $N = 137$: d_{33} reaches a local maximum, shows a slope discontinuity, or exhibits anomalous enhancement relative to the smooth trend from neighboring thicknesses.
- For $N > 137$: d_{33} either saturates or shows a different scaling regime (the additional planes contribute redundant modes, not new ones).

The predicted anomaly magnitude is approximately 0.7% relative to the bulk value (from Toy 1968/1999). This is within the sensitivity of standard PFM measurements (resolution approximately 0.1-0.5%).

2.4 Certificate

Computational certificate (existing). Toy 1967: BaTiO₃ 137-plane Casimir prediction, experimental design (18/18 PASS). Toy 1968/1999: piezoelectric anomaly magnitude calculation. The predicted thickness, anomaly position, and approximate magnitude are all computed.

Experimental certificate (needed). Fabricate 9 films at $N = 129, 131, 133, 135, 137, 139, 141, 143, 145$ unit cells. Measure d_{33} for each. Plot $d_{33}(N)$. Look for a statistically significant anomaly at $N = 137$.

Statistical threshold: the d_{33} value at $N = 137$ deviates from a linear or quadratic fit to its neighbors by more than 2 sigma (measurement uncertainty).

2.5 Boundary (What Is NOT Claimed)

- The prediction applies to the c-axis-oriented tetragonal phase of BaTiO₃ grown epitaxially on SrTiO₃. It does NOT apply to polycrystalline or randomly oriented films, where the thickness-averaged signal smears out single-plane features.
- The prediction is for d_{33} (the longitudinal piezoelectric coefficient). The transverse coefficients d_{31} and d_{15} may or may not show the same anomaly – that is a separate question.
- The prediction does NOT claim that 137 is the global optimum thickness for all applications. It claims an anomalous spectral feature at that thickness.
- Substrate clamping effects from the SrTiO₃ substrate may shift the anomaly by 1-2 unit cells. The prediction is $N = 137 \pm 2$ in the presence of substrate strain.
- The anomaly magnitude (approximately 0.7%) is an order-of-magnitude estimate. A measured anomaly of 0.3-2% at $N = 137$ would be consistent. An anomaly larger than 5% or smaller than 0.1% would require re-examination.

2.6 Cost and Timeline

- Cost: approximately \$25K (PLD target, substrates, deposition time, PFM measurement).
- Timeline: 3-6 months from funding to publication-quality data.
- Where: Any oxide thin film laboratory with PLD capability. Dozens exist worldwide.
- Personnel: One graduate student or postdoc. Standard fabrication and measurement.

2.7 Falsification Criterion

BST fails if: $d_{33}(N)$ shows no statistically significant feature at or within 2 unit cells of $N = 137$, in properly oriented epitaxial c-axis BaTiO₃ films measured with resolution better than 0.5%.

What this would mean: The spectral cap $N_{\text{max}} = 137$ does not manifest in lattice-periodic piezoelectric response. Either N_{max} is not the correct spectral cutoff, or the coupling between BST spectral geometry and condensed matter phonon physics does not work as BST predicts. Either conclusion is fatal to the SE program's core claim.

3. Prediction 2: Photonic Crystal Q-Factor Anomaly at $N = 137$ Periods

The cheapest clean falsification test in the BST program.

3.1 Physical Setup

Fabricate a one-dimensional photonic crystal (Bragg stack) by plasma-enhanced chemical vapor deposition (PECVD) of alternating Si and SiO₂ layers. The standard process produces quarter-wave stacks with layer thicknesses of approximately 100-200 nm, depending on the target bandgap wavelength. Prepare 9 samples with total period counts $N = 125, 128, 131, 134, 137, 140, 143, 146, 149$. Measure the reflectance spectrum of each sample with a standard UV-Vis-NIR spectrophotometer. Extract the quality factor Q of the primary Bragg reflection peak.

3.2 GC Constraint Specification

Constraint 1 (Dielectric periodicity + Bloch's theorem). A periodic dielectric structure with N periods produces a photonic bandgap whose quality factor Q scales with N . For a perfect quarter-wave stack, Q grows exponentially with N up to the point where fabrication imperfections dominate (typically N approximately 50-200, depending on process quality). The dielectric contrast ($n_{\text{Si}}/n_{\text{SiO}_2}$ approximately $3.5/1.46 = 2.4$) determines the gap width; the number of periods determines the gap depth.

Constraint 2 (BST spectral coherence length). The D_{IV}^5 spectral cap $N_{\text{max}} = 137$ sets the maximum number of coherent spectral modes in any periodic structure. In a photonic crystal, each period adds one mode to the Bloch band structure. At $N = N_{\text{max}}$ periods, all independent spectral modes are present. Additional periods beyond N_{max} contribute redundant information – they increase the optical path length but do not add new spectral content.

Independent bounds meet. Constraint 1 (from electromagnetic theory) predicts monotonic Q growth with N . Constraint 2 (from BST) predicts a departure from monotonic behavior at $N = 137$ – either an anomalous enhancement (constructive spectral interference of all 137 modes) or a change in the $Q(N)$ scaling law (crossover from coherent to redundant regime).

3.3 BST-Predicted Outcome

The quality factor $Q(N)$ as a function of period count shows an anomalous feature at $N = 137$:

- $Q(137)$ is approximately 0.7% higher than the value predicted by smooth interpolation from neighboring samples.
- Alternatively, the slope dQ/dN changes character at $N = 137$ (faster growth below, slower above).

The predicted anomaly magnitude is approximately $0.7\% = 1/N_{\text{max}}$ (from Toy 2053). This is small but measurable with high-quality spectrophotometry (typical Q measurement precision is 0.1-0.3%).

3.4 Certificate

Computational certificate (existing). Toy 2053: photonic crystal Q -factor anomaly prediction. 9 samples specified. Standard spectrophotometry protocol.

Experimental certificate (needed). Measure Q for all 9 samples. Fit $Q(N)$ to a smooth function (exponential or polynomial) using the 8 non-137 samples. Compare $Q(137)$ to the fit prediction. Significance threshold: deviation exceeds 2 sigma.

Optionally, supplement with FDTD (finite-difference time-domain) simulation of the same structure. The FDTD simulation uses standard Maxwell's equations with no BST input. If BST is correct, the FDTD simulation should also show the anomaly at $N = 137$, because FDTD faithfully captures the mode structure. If FDTD does NOT show the anomaly, either BST is wrong or the anomaly arises from physics beyond classical electromagnetism (which would itself be interesting).

3.5 Boundary (What Is NOT Claimed)

- The prediction applies to 1D Bragg stacks with high-contrast dielectric pairs (Si/SiO₂, TiO₂/SiO₂, or similar). It does NOT apply to low-contrast stacks where the coherence length is shorter than 137 periods and fabrication imperfections dominate before the BST signal appears.
- The prediction is for the Q of the primary bandgap, not for higher-order gaps or defect modes.

- 2D and 3D photonic crystals may show analogous anomalies, but the geometry is more complex and the prediction less sharp. The 1D case is the clean test.
- The prediction does NOT claim that 137 periods is the optimal engineering design for all photonic crystals. It claims a spectral anomaly at that period count.

3.6 Cost and Timeline

- Cost: approximately \$10K (PECVD deposition, substrates, spectrophotometry).
- Timeline: 3-5 weeks from substrate to data. PECVD is fast.
- Where: Any photonics fabrication facility. Hundreds exist worldwide.
- Personnel: One technician. Completely routine fabrication and measurement.

3.7 Falsification Criterion

BST fails if: $Q(N)$ shows no statistically significant anomaly at or within 2 periods of $N = 137$, in high-quality Si/SiO₂ Bragg stacks with fabrication precision better than 1% layer thickness uniformity.

What this would mean: The spectral cap $N_{\max} = 137$ does not affect the coherence properties of periodic electromagnetic structures. The BST claim that $N_{\max}$ sets a universal spectral coherence length is wrong.

4. Prediction 3: Casimir Heat Engine at Optimal Stroke Ratio $g/\text{rank} = 7/2$

Patent filed April 2, 2026. The BST-unique engineering prediction.

4.1 Physical Setup

Construct a Casimir heat engine that cycles between two plate separations: $d_{\min}$ and $d_{\max}$. The working principle: parallel conducting plates attract via the Casimir force (vacuum mode exclusion). A ferroelectric coating (BaTiO₃) switches between high-permittivity (attractive) and low-permittivity (Lifshitz repulsive) states, enabling a thermodynamic cycle: compress under Casimir attraction, switch to repulsive, expand, switch back. Net work is extracted per cycle.

The BST prediction concerns the optimal stroke ratio $d_{\max}/d_{\min} = g/\text{rank} = 7/2 = 3.5$. This is the ratio of maximum to minimum plate separation that maximizes the work per cycle. The prediction also sets the efficiency bound: $\eta = n_C/g = 5/7 = 71.4\%$.

4.2 GC Constraint Specification

Constraint 1 (Casimir force scaling). The Casimir force between parallel plates scales as F proportional to $1/d^4$. The work extracted in compression from $d_{\max}$ to $d_{\min}$ is W_{compress} proportional to integral of $F \cdot dd$ from $d_{\max}$ to $d_{\min}$, which scales as $1/d_{\min}^3 - 1/d_{\max}^3$. The expansion work (Lifshitz repulsion) scales differently because the repulsive force has a different functional form. The net work $W_{\text{net}} = W_{\text{compress}} - W_{\text{expand}}$ is maximized at a specific $d_{\max}/d_{\min}$ ratio that depends on the force law exponents.

Constraint 2 (BST spectral geometry). The D_{IV}^5 eigenvalue structure constrains the Casimir mode sum through the spectral weights $d(k)$. The asymmetric spacing ratio that maximizes spectral leverage – the ratio at which the attractive and repulsive mode sums have maximum imbalance – is $g/\text{rank} =$

$7/2 = 3.5$. This follows from the BST spectral weight structure: $d(1) = g = 7$ (attractive modes), and the first Wallach point is at rank = 2 (repulsive threshold). The ratio g/rank is the spectral asymmetry of D_{IV}^5 , not a fitted parameter.

Constraint 3 (Efficiency bound). The maximum thermodynamic efficiency of any Casimir cycle is bounded by the ratio of attractive to total spectral modes. BST predicts $\eta_{\text{max}} = n_C/g = 5/7$. This is analogous to a Carnot bound but derived from spectral geometry rather than temperature. The bound is independent of material choice, plate composition, or operating temperature.

Independent bounds meet. Constraints 1 and 2 independently specify the optimal stroke ratio. Constraint 1 comes from the $d^{\{-4\}}$ force law (standard QED). Constraint 2 comes from BST spectral geometry. If BST is correct, the stroke ratio that maximizes W_{net} in a real device is 3.5, not some other number. This is a sharp numerical prediction. Constraint 3 independently bounds the efficiency of the optimal cycle.

4.3 BST-Predicted Outcome

Parameter	BST prediction	BST origin
Optimal stroke ratio $d_{\text{max}}/d_{\text{min}}$	3.5	$g/\text{rank} = 7/2$
Maximum efficiency	71.4%	$n_C/g = 5/7$
Lifshitz repulsion fraction	28.6%	$\text{rank}/g = 2/7$
BaTiO ₃ switching ratio	5.0	n_C (exact)
$\epsilon_{\text{ferro}}/\epsilon_{\text{para}}$		
Optimal absolute gap d_0	137 lattice planes = 54.9 nm (BaTiO ₃)	N_{max}

Each of these is a falsifiable number derived from the five BST integers. No other theory predicts any of them.

4.4 Certificate

Computational certificate (existing). Toys 914, 915, 918, 922: Casimir heat engine cycle analysis. Paper #26 draft v1.1. Toy 1979: Casimir pressure BST scan (13/13 PASS). All BST-predicted parameters verified computationally.

Experimental certificate (needed). Two tiers:

Tier A – force measurement (\$50-100K, 6-12 months). Fabricate a MEMS Casimir device with variable plate separation. Measure the Casimir force as a function of d . Vary $d_{\text{max}}/d_{\text{min}}$ from 2.0 to 5.0 in steps of 0.25. At each ratio, cycle the device and measure net work per cycle. Plot $W_{\text{net}}(d_{\text{max}}/d_{\text{min}})$. BST predicts the maximum is at 3.5.

Tier B – efficiency measurement (\$200-500K, 1-2 years). Build a full Casimir heat engine with BaTiO₃ switching surfaces. Measure the thermodynamic efficiency over many cycles. BST predicts $\eta_{\text{max}} = 71.4\%$. Any measured efficiency exceeding 71.4% would falsify BST.

4.5 Boundary (What Is NOT Claimed)

- The prediction applies to the ideal Casimir cycle with perfect switching and zero friction. Real devices will have lower efficiency due to switching losses, mechanical friction, and thermal noise.

BST predicts the BOUND, not the achieved efficiency.

- The optimal stroke ratio 3.5 is for parallel-plate geometry. Curved or corrugated geometries may have different optima (the geometry changes the mode sum). The prediction is for flat plates.
- The efficiency bound $\eta = 5/7$ applies to any Casimir cycle, regardless of material choice. But the optimal gap $d_0 = 137$ planes is specific to BaTiO₃. Other materials will have different optimal gaps ($N_{\max}$ times their lattice constant).
- The prediction does NOT claim that the Casimir heat engine violates thermodynamics. The energy source is vacuum fluctuations, which have nonzero energy density. The engine converts vacuum energy to mechanical work, bounded by $\eta = 5/7$, which is less than 1. No perpetual motion is claimed.

4.6 Cost and Timeline

- Cost: Tier A force measurement approximately \$50-100K; Tier B efficiency measurement approximately \$200-500K.
- Timeline: Tier A 6-12 months; Tier B 1-2 years.
- Where: MEMS fabrication facility with Casimir measurement capability. Groups at Yale (Lamoreaux), Purdue (Iannuzzi), NIST, and Padova have the equipment.
- Patent status: Provisional filed April 2, 2026.

4.7 Falsification Criterion

BST fails if: (a) the net work per cycle $W_{\text{net}}(d_{\max}/d_{\min})$ does not peak at 3.5 ± 0.3 , or (b) a measured efficiency exceeds 71.4% in any Casimir cycle, or (c) the BaTiO₃ dielectric switching ratio is not 5.0 ± 0.5 .

What this would mean: The spectral asymmetry of D_{IV}^5 ($g/\text{rank} = 7/2$) does not govern the Casimir mode sum in real devices. Either the BST spectral weights are wrong, or the coupling between D_{IV}^5 spectral geometry and Casimir physics does not work as predicted. Either conclusion falsifies the SE program's core engineering claims.

5. Comparison Table: Traditional vs. GC-Formalized Engineering

Aspect	Traditional approach	GC-formalized approach
Design parameter selection	Empirical optimization, parameter sweeps, trial and error	Independent constraints pin the optimal parameters – no sweep needed
Predicted outcome	“We expect improved performance in this range”	“The anomaly is at $N = 137 \pm 2$, magnitude approximately 0.7%”
Optimality claim	“We found the best performance at this setting”	“The optimal ratio is $g/\text{rank} = 3.5$ – uniquely forced by spectral geometry”
Falsifiability	Weak – negative result means “try different parameters”	Strong – negative result kills a specific theoretical claim
Cross-field connection	None – each experiment is self-contained	All three experiments test the same spectral cap ($N_{\max} = 137$) via different physics

Aspect	Traditional approach	GC-formalized approach
Scope limitation	Often unstated or buried in supplementary material	Explicit boundary statement: what is NOT claimed
Efficiency bounds	Material-dependent, empirical	Universal bound from geometry ($\eta = 5/7$ for ANY Casimir cycle)
Number of free parameters	Typically several (material properties, geometry, operating conditions)	Zero – all parameters derived from five integers

The key difference: traditional engineering optimizes within a parameter space. GC-formalized engineering identifies the unique point in parameter space forced by independent constraints. The traditional approach asks “what works best?” The GC approach asks “what is the only thing that can work?” When the answer exists, GC eliminates the parameter sweep entirely.

6. Connection to GC-11: From General Pattern to Specific Tests

GC-11 (Engineering Applications of Geometric Constraint) catalogs eight fields where geometric constraint is already the operative mechanism. Three of those fields – photonic crystals, superconductor design, and crystal structure prediction – have direct BST connections. This note takes the next step: for each BST-connected field, we specify the exact experiment that tests whether BST’s spectral geometry adds predictive power beyond the standard engineering framework.

GC-11 field	GC-11 assessment	GC-12 specific test	BST integer tested
Photonic crystals	“Moderate GC value-add”	Prediction 2: Q-factor anomaly at $N = 137$	$N_{\max}$
Superconductor design	“Conditional on BST verification”	(Future: B12H32 synthesis, pred_101)	Multiple
Crystal structure prediction	“High GC value-add”	Prediction 1: BaTiO ₃ 137-plane anomaly	$N_{\max}$
Metamaterials	“Moderate”	Prediction 3: Casimir engine at $g/\text{rank} = 3.5$	g, rank

GC-11’s honest assessment was that GC “adds the most value where the classification is discrete and finite, the exclusion step is under-formalized, and cross-field connections are non-obvious.” All three GC-12 predictions satisfy these criteria:

- Discrete classification: The film thickness N is an integer. The period count is an integer. The stroke ratio is a rational number. The candidate space is countable.
- Exclusion under-formalized: Standard engineering does not explain why $N = 137$ should be special. BST does. If the prediction holds, BST adds structure that was invisible before.
- Cross-field connection: All three experiments test the same underlying spectral cap ($N_{\max} = 137$) in three different physical systems (piezoelectric, photonic, vacuum). A positive result in any one experiment supports the claim that $N_{\max}$ is universal. A positive result in all three would be strong evidence that D_{IV}^5 spectral geometry governs condensed matter physics.

7. Why Falsifiability Matters

7.1 Each experiment can kill BST

This is not rhetoric. It is the design criterion.

- If BaTiO₃ thin films show no anomaly at 137 planes, BST's spectral cap does not manifest in piezoelectric physics. The SE program's flagship prediction fails.
- If photonic crystal Q-factors show no anomaly at 137 periods, BST's spectral coherence length is wrong. The claim that $N_{\max}$ sets a universal bound on periodic structure performance is falsified.
- If the Casimir engine's optimal stroke ratio is not 3.5, BST's spectral asymmetry (g/rank) does not govern vacuum physics. The five integers do not control Casimir engineering.

Any one of these failures would require fundamental revision of BST's claim that D_{IV}^5 spectral geometry is the source of physical law. Not a patch, not a correction term – a fundamental rethinking of whether the spectral projection mechanism works as claimed.

7.2 Why BST invites this risk

Most theories avoid sharp falsifiable predictions because failure is expensive (reputationally, not financially). BST invites them because:

1. 600+ post-dictions at <1% precision. The theory has earned the right to make pre-dictions. Demanding falsifiable tests is what distinguishes science from numerology.
2. The predictions cost almost nothing. \$10K for the photonic crystal, \$25K for the BaTiO₃ film, \$50-100K for the Casimir force measurement. These are graduate-student-scale experiments, not billion-dollar colliders.
3. A null result is still valuable. If BST is wrong about $N_{\max} = 137$ in condensed matter, that is important information. It would mean the spectral cap applies only to particle physics (where it gives $\alpha = 1/137$), not to all periodic structures. That boundary would itself be a discovery.

7.3 The three-prediction correlated test

The strongest version of the falsification protocol runs all three experiments. If BST is correct, all three show anomalies at $N_{\max}$ -related values. If BST is wrong, most likely none do (the spectral cap mechanism fails uniformly). The interesting case is partial success – anomaly in one or two experiments but not the third. This would indicate that the spectral cap couples to some physical systems but not others, which would narrow the scope of BST's SE claims without killing the theory entirely. The boundary statement for partial success is: “ $N_{\max}$ governs [the systems where the anomaly appears] but not [the system where it does not].”

8. Summary

#	Prediction	BST integer	Cost	Timeline	Kill criterion
1	BaTiO ₃ d_{33} anomaly at 137 planes	$N_{\max} = 137$	\$25K	3-6 months	No anomaly at $N = 137 \pm 2$

#	Prediction	BST integer	Cost	Timeline	Kill criterion
2	Photonic crystal Q anomaly at 137 periods	N_max = 137	\$10K	3-5 weeks	No anomaly at N = 137 +/- 2
3	Casimir engine optimal stroke ratio 3.5	g/rank = 7/2	\$50-100K	6-12 months	Peak not at 3.5 +/- 0.3

Total cost for all three experiments: approximately \$85-125K.

Total time to first results: 3-5 weeks (photonic crystal test alone).

If all three succeed, BST's spectral engineering program is empirically grounded. If any one fails, BST has a specific, identified problem to address or acknowledge. That is how science works.

References

- GC-5: notes/BST_GC5_Five_Step_Methodology.md (Three-move and five-step methodology)
- GC-11: notes/BST_GC11_Engineering_Applications.md (Engineering applications survey)
- Substrate Engineering Priorities: notes/BST_Substrate_Engineering_Priorities.md (Priority pyramid)
- Spectral Engineering Investigation: notes/BST_Spectral_Engineering_Investigation.md (Full SE plan)
- Patentable Applications Catalog: notes/BST_Patentable_Applications_Catalog.md (25 devices)
- pred_096: BaTiO3 137-plane prediction (data/bst_predictions.json)
- pred_097: Photonic crystal Q-factor prediction (data/bst_predictions.json)
- Toys 914, 915, 918, 922: Casimir heat engine cycle analysis
- Toy 1967: BaTiO3 137-plane experimental design (18/18 PASS)
- Toy 1968/1999: Piezoelectric anomaly magnitude
- Toy 1979: Casimir pressure BST scan (13/13 PASS)
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GC-12 converts GC-11's general survey into specific falsifiable tests. Three experiments, three BST integers, three ways to kill BST. Total cost: \$85-125K. First result: 3-5 weeks. The math makes predictions. The lab decides.